Credit Risk Portfolio Analytics

Product brief | Prototype for portfolio monitoring and analyst decision support

Problem

Credit-risk teams need a consolidated view of customer, loan-account, bureau, repayment and collections data to spot portfolio concentrations, understand risk patterns, review unusual profiles and evaluate whether risk scores are useful. In fragmented data, analysts can spend time reconciling tables and may miss emerging concentrations or data-quality problems. This prototype brings the five synthetic datasets into one interactive workspace.

Target users

Primary: portfolio risk analysts, credit-risk managers and model-risk reviewers. Secondary: collections strategy, data/BI teams and a Chief Risk Officer who needs a concise view of portfolio composition, model discrimination and exceptions. It is a decision-support tool—not an automated underwriting system.

Scope — in

- Load and validate five linked CSV tables; report missingness, duplicate/key issues and documented assumptions.
- Portfolio KPIs and interactive cuts across available customer/account dimensions, with account-level exploration.
- Benchmark Logistic Regression and XGBoost using AUC, KS, Gini, Brier score and threshold metrics; show feature importance and evaluation visuals where available.
- Isolation Forest anomaly screening on selected origination/profile variables, with adjustable review threshold and CSV export.
- Ask the Portfolio: local Ollama-powered explanations over allow-listed, verified Python calculations; numeric results are calculated by the app, not invented by the LLM.
- A monitoring plan covering PSI/CSI, matured-label performance, data quality, drift investigation, human review and retraining governance.

Scope — out

- Production credit decisioning, automated approve/decline, pricing or limit setting.
- Live bank-system integration, real-time scoring, customer-facing use or SLA-backed service.
- Use of the current synthetic data as evidence of real-bank performance or regulatory fitness.
- Causal claims, forecasts, fraud conclusions or time-trend claims not supported by validated data.
- Fully validated SHAP explanations, formal fairness testing, independent model validation, production security and regulatory sign-off.

Success criteria for a pilot

A risk analyst can reproduce portfolio cuts, inspect flagged profiles, trace metrics to a defined evaluation set, understand key limitations and export a review list. Before production, the bank must approve target definition, data lineage, access controls, model validation, monitoring thresholds and human-review procedures.